

Lecture # 14

Date: Oct 7th, 25

Day: Tuesday

CFG $\rightarrow$ CNF

$\hookrightarrow \begin{cases} S \rightarrow \Lambda / VV \\ V \rightarrow VV / \Sigma \end{cases}$

$S \rightarrow \overline{VV} ASB / \Lambda \Sigma VV$

$A \rightarrow aAS / a / \Lambda, \Sigma$

$B \rightarrow \overline{VS} bS / A / \overline{\Sigma\Sigma} bb$

$\rightarrow$ intermediate level pe 'A'

Steps:

1. Make a new start state :

$S_n \rightarrow S$

$S \rightarrow ASB / \Lambda$

$A \rightarrow aAS / a / \Lambda$

$B \rightarrow SbS / A / bb / V$

2. Eliminate null productions:

($S \rightarrow \Lambda$)

$S_n \rightarrow S / \Lambda$

$S \rightarrow ASB / AB$

$A \rightarrow \overline{aAS} aAS / a / \Lambda / aA$

$B \rightarrow SbS / A / bb / b / Sb / bs$

($A \rightarrow \Lambda$)

$S_n \rightarrow S / \Lambda$

$S \rightarrow ASB / AB / \overline{SB} B$

$A \rightarrow aAS / a / aA / aS$

$B \rightarrow SbS / A / bb / b / Sb / bs / \Lambda$

($B \rightarrow \Lambda$)

$S_n \rightarrow S / \Lambda$

$S \rightarrow ASB / AB / SB / B / AS / A / S / \Lambda$

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$$A \rightarrow aAS | a | aA | aS$$

$$B \rightarrow SbS | A | bb | b | Sb | bS |$$

$(S \rightarrow \Lambda)$ ki removal ---- Already checked ---
no change in overall grammar.

Step 3: Remove unit production.

$$VA \rightarrow VA \quad \text{or} \quad VA \rightarrow VB$$

$$\bullet VA \rightarrow VB$$

$$\bullet VA \rightarrow VA$$

$$SN \rightarrow S$$

$$S \rightarrow S$$

→ self calls directly

$$S \rightarrow A$$

eliminate ---- copy the

$$S \rightarrow B$$

same rules (as redundant)

$$B \rightarrow A$$

already present

$$SN \rightarrow ASB | AB | SB | AS | SbS | aAS | a | aA | aS | bb | b | Sb | bS | \Lambda$$

$$S \rightarrow ASB | AB | SB | AS | SbS | aAS | a | aA | aS | bb | b | Sb | bS$$

$$A \rightarrow aAS | a | aA | aS$$

$$B \rightarrow SbS | \underbrace{aAS | a | aA | aS}_A | bb | b | Sb | bS$$

Step 4: Transformation (introduce new variables)

$$P \rightarrow a$$

$$Q \rightarrow b$$

$$SN \rightarrow ASB | AB | SB | AS | SQS | PAS | a | PA | PS | QQ | b | SQ | QS | \Lambda$$

$$S \rightarrow ASB | AB | SB | AS | SQS | PAS | a | PA | PS | QQ | b | SQ | QS$$

$$A \rightarrow PAS | a | PA | PS$$

$$B \rightarrow SQS | PAS | a | PA | PS | QQ | b | SQ | QS$$

out of format

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$R \rightarrow AS$

$S_N \rightarrow RB|AB|SB|AS|SQS|PR|a|PA|PS|QQ|b|SQ|QS|$
 $S \rightarrow RB|AB|SB|AS|SQS|PR|a|PA|PS|QQ|b|QS|SQ$
 $A \rightarrow PR|a|PA|PS$
 $B \rightarrow SQS|PR|a|PA|PS|QQ|b|SQ|QS$

$W \rightarrow QS$

$S_N \rightarrow RB|AB|SB|AS|SW|PR|a|PA|PS|QQ|b|SQ|QS|$
 $S \rightarrow RB|AB|SB|AS|SW|PR|a|PA|PS|QQ|b|QS|SQ$
 $A \rightarrow PR|a|PA|PS$
 $B \rightarrow SW|PR|a|PA|PS|QQ|b|SQ|QS.$

$W \rightarrow QS, R \rightarrow AS, P \rightarrow a, Q \rightarrow b$

Practice Ques:

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$$A \rightarrow BAB|B|\Lambda$$

$$B \rightarrow 00|\Lambda$$

$$S_v = \{A\}$$

$$V = \{A, B\}$$

$$\Sigma = \{0\}$$

Step I:

$$S_n \rightarrow A$$

$$A \rightarrow BAB|B|\Lambda$$

$$B \rightarrow 00|\Lambda$$

Step II:
(A $\rightarrow \Lambda$)

$$S_n \rightarrow A|\Lambda$$

$$A \rightarrow BAB|B|BB$$

$$B \rightarrow 00|\Lambda$$

(B $\rightarrow \Lambda$)

$$S_n \rightarrow A|\Lambda$$

$$A \rightarrow BAB|B|BB|AB|BA|A$$

$$B \rightarrow 00|\Lambda$$

already removed

Step III:

$VA \rightarrow VB$: $S_n \rightarrow A, A \rightarrow B$

$$S_n \rightarrow BAB|00|BB|BA|AB|\Lambda$$

$$A \rightarrow BAB|00|BB|AB|BA|A$$

$$B \rightarrow 00$$

$V_A \rightarrow V_n$

Step IV:

let $P \rightarrow 0$

$$S_n \rightarrow BAB|PP|BB|BA|AB|\Lambda$$

$$A \rightarrow BAB|PP|BB|AB|BA$$

$$B \rightarrow PP$$

$Q \rightarrow BA$

$$S_n \rightarrow QB|PP|BB|BA|AB|\Lambda$$

$$A \rightarrow QB|PP|BB|BA|AB$$

$$B \rightarrow PP$$

Lecture #15

Date Oct 9th, 2025

Day: Thursday

$$S \rightarrow aB \mid bA \mid \Lambda$$

$$A \rightarrow aS \mid bAA$$

$$B \rightarrow bS \mid aBB$$

1:

$$S_n \rightarrow S$$

2:

$$S_n \rightarrow S \mid \Lambda$$

$$S \rightarrow aB \mid bA \mid \cdot$$

$$A \rightarrow aS \mid bAA \mid \text{~~aa~~} \mid \text{~~a~~} \mid a$$

$$B \rightarrow bS \mid aBB \mid \text{~~ab~~} \mid \text{~~b~~}$$

3:

$$S_n \rightarrow S$$

$$S_n \rightarrow aB \mid bA \mid \Lambda$$

$$A \rightarrow aS \mid bAA \mid \text{~~aa~~} \mid a$$

$$B \rightarrow bS \mid aBB \mid \text{~~ab~~}$$

$\forall$
 Σ

4. $Q \rightarrow b$

$$P \rightarrow a$$

$$S_n \rightarrow PB \mid QA \mid \Lambda$$

$$S \rightarrow PB \mid QA$$

$$A \rightarrow PS \mid QAA \mid \text{~~Qa~~}$$

$$B \rightarrow QS \mid PBB \mid b$$

$$R \rightarrow QA$$

$$T \rightarrow PB$$

$$S_n \rightarrow PB \mid QA \mid \Lambda$$

$$S \rightarrow PB \mid QA$$

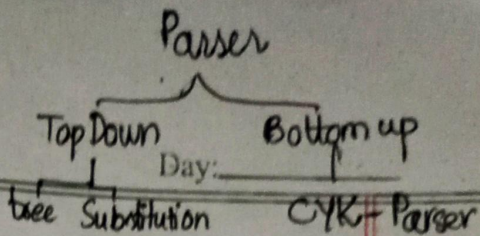
$$A \rightarrow PS \mid RA \mid a$$

$$B \rightarrow QS \mid TB \mid b$$

Date:

check membership!

CYK - Parser



$\Sigma^* \Rightarrow S_N$

$\Rightarrow S_N$ string belongs to Grammar.

=4	1	{S _n , S, T}	abba	X	X	X						
=3	2	{B}	abb	3	{B}	bba	X					
=2	4	{S _n , S, T}	ab	5	{ }	bb	6	{S _n , S, R}	ba	X		
j=1	7	{A, P}	a	8	{B, Q}	b	9	{B, Q}	b	10	{A, P}	a
			a		b		b		a			

{A, P} * {B, Q}

cartesian prod.

AB → x

AQ → x

PB {S_n, S, T}

PQ → x

abba
abba abb a a-bba

{B, Q} * {B, Q}

{BB, BQ, QB, QQ}

{B, Q} * {A, P}

BA, BP, QA, QP

2, 5, 14

S_n, S, R.

{B, Q} {S_n, S, R}

BS, BR

QS, QR

5, 6

abb: 4, 9

{S_n, S, T} {B, Q}

bba
b-ba bb-a
8, 6 5, 10

$\{B\}$ ^{abbab} abba abb $\{B\}$	$\{ \}$ ^{bbab} bba $\{B, \}$ ^{ba} $\{S_n, S, R\}$ "	^{bab} $\{B, \}$ ^{ba} $\{S_n, S, T\}$ "	^{ab} $\{S_n, S, T\}$ "	$\times$ $\{B, Q\}$
a	b	b	a	b

bab

bab

$\overline{abbab}$